STUDIFYX: AN INTEGRATED EXPLAINABLE AI PLATFORM FOR ACADEMIC GUIDANCE, CAREERS AND WELLBEING

INDIVIDUAL THESIS

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ABSTRACT

Students often finish their degrees with little access to people who can explain how careers in their field work. Mentoring platforms try to close this gap, but most rely on self-declared profiles, so a student cannot tell whether a mentor's credentials are genuine, and manual vetting does not scale. This thesis presents the Professional Mentorship and Alumni Network component of StudiFyx, an AI-driven educational platform. The component connects students with mentors from universities and industry and verifies those mentors through an automated, layered trust chain.

The research problem is how to establish mentor credibility without manual validation and without keeping sensitive documents. The component combines a matching model based on academic interests, degree programme and career goals; document verification that reads curricula vitae (CVs) and certificates with optical character recognition (OCR); a structured interview that is scored automatically; Bronze, Silver and Gold verified badges derived from the combined verification strength; a secure real-time chat; and a privacy layer that holds documents only in temporary encrypted storage, keeps the extracted metadata and deletes the raw files automatically. The module is integrated with the central intelligence application programming interface (API) of the platform, so that human mentorship acts as a decision-support layer beside the AI recommendations.

The component was built with React and Node.js and is evaluated on field-extraction accuracy, agreement with a human reviewer, matching relevance, chat latency, deletion compliance and usability. The contributions are a multi-layered trust-chain verification method, a privacy-preserving credibility check, a graded Trusted Mentor badge, and the formal integration of mentorship into AI-based guidance.

Keywords: mentorship platform, mentor verification, OCR, privacy-preserving verification, trust badges

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AES	Advanced Encryption Standard
AI	Artificial Intelligence
API	Application Programming Interface
CV	Curriculum Vitae
F1	Harmonic mean of precision and recall
FR	Functional Requirement
GCM	Galois/Counter Mode
H	Hypothesis
HTTP	Hypertext Transfer Protocol
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
JSON	JavaScript Object Notation
JWT	JSON Web Token
nDCG	Normalised Discounted Cumulative Gain
NFR	Non-Functional Requirement
OCR	Optical Character Recognition
OWASP	Open Worldwide Application Security Project
REST	Representational State Transfer
RQ	Research Question
SUS	System Usability Scale
TLS	Transport Layer Security

1 INTRODUCTION

1.1 Overall project context

StudiFyx is an AI-driven educational platform that helps students make study and career decisions. Its features are built as separate components that share data and services through a central intelligence application programming interface (API). Each component was built by a different team member, and each member reports on their own component in an individual thesis. This thesis reports only on the Professional Mentorship and Alumni Network component, which I developed as Member 4. Other components appear only where the mentorship component exchanges information with them through the central intelligence API.

The platform's AI features can suggest courses and career directions, but a suggestion from software is not the same as advice from someone who has done the job. The mentorship component adds that human layer: a student can find a mentor, see how strongly that mentor has been verified, and talk to them in real time.

1.2 Individual component context

The component is a mentorship module made of a React front end and a Node.js back end. The mentor record is defined in `src/members/member4.jsx` under the component ID `member4`. It holds the identity fields (`id`, `name`, `email`, `role`, `dashboard`) and a nested `profile` with first name, last name, biography, expertise list, courses created, students mentored and average rating. This record is the data foundation of the module: the matching logic reads the expertise list, the dashboard reads the profile, and the verification service writes its results into the same record.

The component has six functions. It matches students to mentors using academic interests, degree programme and career goals. It verifies mentors by reading their CVs and certificates with optical character recognition (OCR) and by scoring a structured interview. It provides a real-time chat between a student and a mentor. It shows mentor profiles with Bronze, Silver or Gold verified badges and with visibility settings that the mentor controls. It protects the uploaded documents by keeping them only briefly, encrypted, and deleting them once their metadata has been extracted.

Finally, it exposes these functions to the central intelligence API so that the rest of the platform can use them.

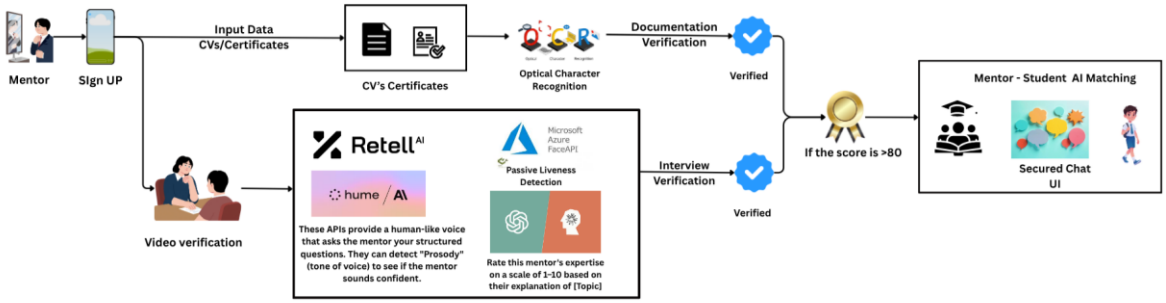


Figure 1.1: Overview of the Professional Mentorship and Alumni Network component

1.3 Problem statement

Students who are about to enter the workplace often lack practical knowledge of how careers in their field develop. Research on mentoring shows that people who have mentors report better career outcomes than those who do not [1], [2]. A mentor can close this gap only if the student can trust that the mentor is who they claim to be and has the experience they describe. On open platforms a mentor profile is usually self-declared, and a student has no easy way to check it.

Checking mentors by hand does not scale. A reviewer has to read each CV, compare it with the certificates and speak to the applicant. Curated mentoring marketplaces accept this cost by screening applicants carefully and admitting only a small share of them [3]. A university platform that wants to bring many alumni on board cannot depend on that approach.

Verification also creates a privacy problem. A CV and a certificate contain a person's full name, dates, employment history and sometimes identity numbers. Keeping these files permanently to prove that a check took place turns the platform into a target for attackers and conflicts with the data minimisation and storage limitation principles of data protection law [4], [5]. The problem addressed here is therefore: *how can a platform establish and communicate the credibility of mentors*

automatically, in several layers, without a manual reviewer and without keeping the sensitive documents it examines?

1.4 Research gap

The literature and public documentation on mentoring platforms point to four gaps that this component addresses.

- **Verification depends on self-declaration or manual review.** Open profiles are not checked, and curated services use human screening that limits how many mentors can join [3].
- **Document checks keep the documents.** Systems that read CVs and certificates are usually built for hiring or onboarding and store the source files. Few state how long a document stays or what remains after processing.
- **Trust signals are coarse.** A mentor is typically verified or not, or judged by reviews written later. A graded signal that reflects how much evidence was checked is uncommon.
- **Mentoring sits beside AI guidance, not inside it.** Human mentorship is rarely a formal input to an AI decision-support platform.

1.5 Aim and objectives

Aim. To design, implement and evaluate a mentorship and alumni-network component that connects students with mentors who are verified automatically through documents and an interview, while keeping the documents themselves private.

Objectives. Six objectives follow the tasks assigned to this component.

O1 Design mentor–student matching logic based on academic interests, degree programmes and career goals.

O2 Implement automated mentor verification through CV and certificate upload with OCR-based text extraction, and a structured interview (questionnaire or short video) with scoring.

O3 Develop a secure real-time chat channel between students and mentors.

O4 Create mentor profiles with Bronze, Silver and Gold verified badges and controlled visibility settings.

O5 Make verification privacy-preserving through temporary encrypted storage, metadata-only extraction and automatic deletion of raw files.

O6 Integrate the mentorship module with the central intelligence API.

1.6 Research questions or hypotheses

The study is guided by four research questions (RQ).

RQ1 How accurately can OCR-based extraction read the fields needed for credential checks (name, institution, degree, year, employer and job title) from CVs and certificates?

RQ2 How closely do the automated badge decisions agree with the decisions of an independent human reviewer?

RQ3 Does the matching score rank mentors in an order that students judge to be relevant?

RQ4 Can verification be completed so that no raw document remains in storage after processing, without making the process too slow to use?

Three hypotheses (H) are tested. **H1:** field-level F1 (the harmonic mean of precision and recall) is at least 0.85 on documents of acceptable scan quality. **H2:** agreement between automated and reviewer badge decisions reaches Cohen's kappa of at least 0.61. **H3:** no raw document file remains after the retention window in any test run.

1.7 Scope limitations and assumptions

In scope. Mentor profiles, document and interview verification, badge assignment, matching, mentor-controlled visibility, real-time chat, privacy-preserving document handling and the interface to the central intelligence API.

Out of scope. Course authoring (the `coursesCreated` field is kept in the data model but no authoring tools are built), session booking and calendar scheduling, audio or video calls, payments, and authenticity checks against the issuing institutions.

Assumptions and limitations. Uploaded documents are in English and are PDF, JPEG or PNG files. Users are authenticated by the platform's existing login, and student interests, programme and career goals come from the student profile. OCR and consistency checking show that a document agrees with a mentor's claims; they do not prove that the document is genuine (Section 7.5).

1.8 Individual contribution

My individual contribution is the design, implementation and evaluation of the complete mentorship component. Table 1.1 maps each assigned task to what I built and to the section where it is described.

Table 1.1: Assigned tasks and individual contribution

Task	Contribution	Section
Mentor–student matching	Weighted content-based score from interests, degree programme, career goals and badge level	4.4, 5.2
Automated mentor verification	OCR extraction from CVs and certificates, document score, scored structured interview	4.4, 5.2
Real-time communication	Authenticated chat limited to accepted mentorships	4.4, 5.2
Verified mentor profiles	Bronze, Silver and Gold badges and mentor-controlled visibility	4.4, 5.2
Privacy-preserving verification	Advanced Encryption Standard in Galois/Counter Mode (AES-256-GCM) temporary storage, metadata-only retention, automatic deletion	4.4, 5.2
Platform integration	Versioned Representational State Transfer (REST) endpoints for the central intelligence API	5.3

The work is novel in four ways.

- A multi-layered trust chain that verifies mentors through OCR-based document analysis and interview scoring and removes manual validation from the verification path.
- A privacy-preserving credibility check that never stores sensitive documents or credentials permanently.
- A Trusted Mentor badge system with Bronze, Silver and Gold levels that tells students how strongly a mentor has been verified.

- The integration of human mentorship as a formal decision-support layer inside the AI platform.

1.9 Thesis structure

Chapter 2 reviews the literature, Chapter 3 the methodology, and Chapter 4 the requirements and design. Chapter 5 covers implementation, integration and testing. Chapters 6 to 8 give the results, discussion and conclusion. The appendices hold the interview items and the AI-use disclosure.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The review covers the five areas on which the component depends: mentoring research, trust and credential verification, text extraction from documents, recommendation and matching, and secure real-time communication. A source was included when it informed a specific design decision in Chapter 4. Standards and legal texts were included where the component has to comply with them. Public documentation of one mentoring marketplace was used for the comparison in Section 2.4.

2.2 Domain and theoretical foundations

Mentoring. Two meta-analyses show that mentoring is associated with better outcomes. Eby et al. found that mentored people reported more favourable behavioural, attitudinal, relational and career outcomes than people without mentors [1]. Allen et al. found that mentored protégés reported higher compensation, more promotions and greater career satisfaction [2]. Kram described mentoring as a relationship that moves through phases, beginning with initiation and then cultivation [6]. The component supports these first phases: finding a suitable mentor and keeping a channel open for regular contact. The handbook literature treats the quality of the match and the credibility of the mentor as central to success [7].

Trust and assurance levels. Digital identity guidance defines assurance levels that rise with the strength of the evidence collected about a person [8]. The Bronze, Silver and Gold badges of this component follow the same idea by analogy: more independent evidence and a higher interview score give a higher level. The badges are not claimed to meet any formal assurance standard.

Privacy and storage. The General Data Protection Regulation requires that personal data be limited to what is necessary and kept no longer than necessary [4]. Sri Lanka's Personal Data Protection Act sets out comparable obligations [5]. For the encrypted temporary store, authenticated encryption with AES in Galois/Counter Mode protects both confidentiality and integrity of the stored bytes [9].

2.3 Related research

Text extraction. Tesseract is an open-source OCR engine with a long development history [10]. Its accuracy depends on image quality, resolution and layout, and it returns plain text, so fields such as degree or employer must be located by further processing. OCR errors (for example a missing letter in a name) mean that extracted values cannot be compared with exact matching. Edit distance [11] gives a measure of how many single-character changes separate two strings and is widely used to tolerate such errors.

Matching and recommendation. Recommender systems are commonly grouped into collaborative, content-based and hybrid approaches [12], [13]. Collaborative methods need a history of ratings, which a new mentorship platform does not have. Content-based methods compare the attributes of the two sides directly and can explain why a pair was chosen. For comparing sets of tags such as interests and expertise, the Jaccard coefficient [14] measures overlap as the size of the intersection divided by the size of the union. Ranked lists are evaluated with precision at k and with normalised discounted cumulative gain (nDCG) [15].

Real-time communication and APIs. The WebSocket protocol provides a persistent two-way channel between browser and server [16]. JSON (JavaScript Object Notation) Web Tokens (JWT) carry signed identity claims that a server can check without a session lookup [17]. The REST architectural style [18] guides the design of the versioned Hypertext Transfer Protocol (HTTP) interface used to integrate with the central intelligence API.

2.4 Existing systems products and techniques

Table 2.1 compares four approaches to mentor credibility. The comparison is qualitative. The marketplace row relies on its public help documentation [3], and the other rows describe typical practice rather than a specific product.

Table 2.1: Approaches to establishing mentor credibility

Approach	How credibility is established	Handling of documents	Main limitation
Open professional or social profiles	Self-declared profile	No documents collected	Credentials cannot be checked by students
Curated mentoring marketplace [3]	Application and screening review; ongoing assessment of activity and feedback	Not described in the cited source	Manual screening limits scale; under 15% of applicants admitted
Institutional alumni portal	Alumni record or institutional email, typically	Varies by institution	Shows membership, not professional experience
Proposed component	Three layers: document evidence, scored interview, graded badge	Encrypted for a short time, metadata kept, raw files deleted	Checks consistency, not authenticity (Section 7.5)

None of the approaches combines automated multi-layer verification with a stated limit on how long documents are kept. The component occupies that space.

2.5 Gap synthesis and conceptual framework

Combining the findings above gives the conceptual framework in Figure 2.1. It rests on three ideas.

- **Evidence in, metadata out.** Documents are used only to produce a small set of extracted fields and confidence values. Once these exist, the document is destroyed.
- **Layered trust.** The document layer shows that the mentor's claims match written evidence. The interview layer shows that the mentor can explain their field and behave well as a mentor. Each layer adds to a combined verification strength.
- **Strength made visible.** The combined strength is shown to students as a Bronze, Silver or Gold badge, and it also controls whether a mentor can be matched at all and how highly they rank.

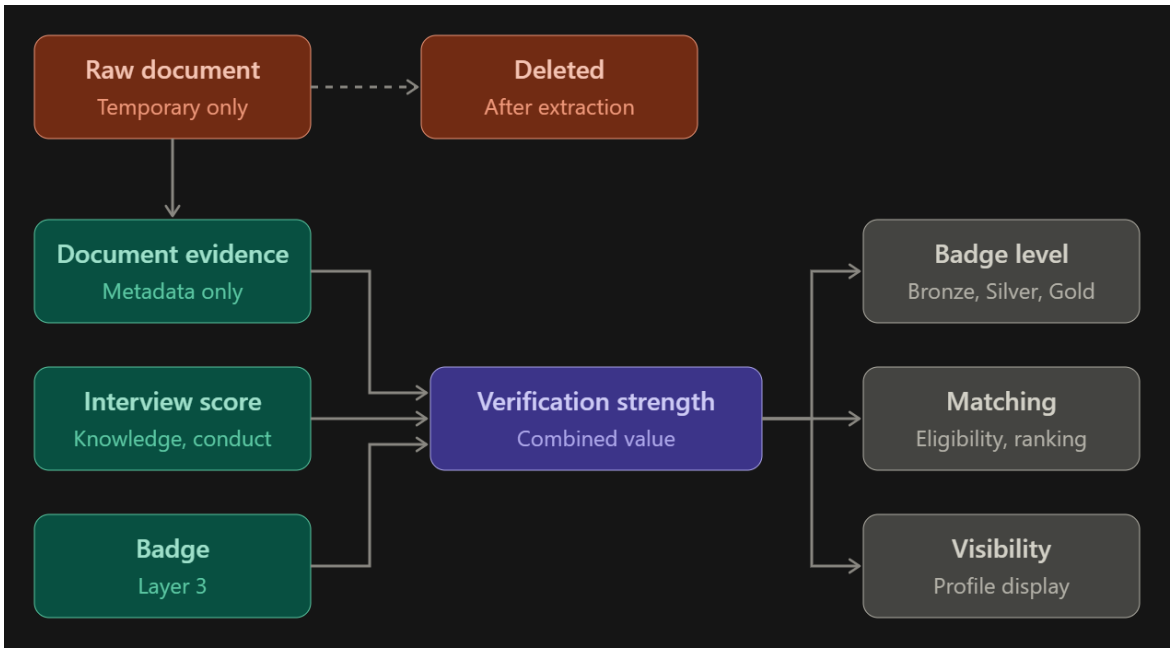


Figure 2.1: Conceptual framework of the trust-chain verification model

3 RESEARCH METHODOLOGY

3.1 Research design

The research follows the design science approach [19]: a problem was identified (Section 1.3), objectives were set (Section 1.5), an artefact was built (Chapters 4 and 5) and evaluated against the objectives (Chapter 6). The work was done in short iterations, each adding one layer or feature and testing it. Evaluation uses quantitative measures for extraction, verification, matching and performance, and user ratings for relevance and usability. Figure 3.1 shows the workflow.

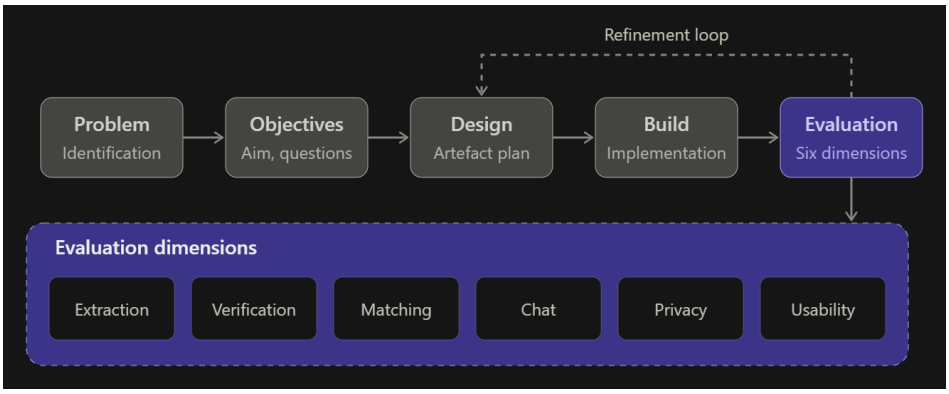


Figure 3.1: Research design and evaluation workflow of the mentorship component

3.2 Data participants or study materials

Table 3.1 lists the study material. No real person's credentials were retained. Test documents were synthetic (invented persons) or supplied by volunteers who agreed to their use.

Table 3.1: Study materials and participants

Material	Description	Planned size
Test documents	CVs and degree or professional certificates in clean PDF, scanned and phone-photo form; synthetic and volunteer-supplied	60 (30 CVs, 30 certificates)
Mentor volunteers	Alumni and industry professionals who completed verification	20
Student participants	Students who stated interests, programme and career goals and rated matches	30
Interview items	Questions in four domains (Appendix A)	12 items
Reference labels	Field values and badge decisions set independently by a human reviewer	All of the above

3.3 Experimental or study procedure

The evaluation has seven steps.

1. A reviewer records the correct value of each field in every test document (ground truth).
2. Each document goes through the OCR pipeline, and extracted fields are compared with the ground truth.
3. Each mentor volunteer uploads documents and completes the interview. The system computes the badge, and the reviewer independently assigns the level the evidence deserves under the written criteria.
4. Each student enters interests, programme and career goals, receives the top five mentors and rates each for relevance from 1 to 5.
5. A load script simulates concurrent chats and records delivery latency.
6. After each verification run and after the retention window, temporary storage and logs are inspected for leftover files or plaintext fragments.
7. Students and mentors complete the System Usability Scale (SUS) after a set of tasks.

3.4 Measures and evaluation metrics

Table 3.2: Evaluation metrics and targets

Metric	Definition	Target
Field precision, recall, F1	Correct extracted fields against ground truth, per field and macro-averaged	F1 ≥ 0.85 (H1)
Cohen's kappa	Agreement between automated and reviewer badge levels, corrected for chance	$\kappa \geq 0.61$ (H2)
False accept rate	Share of mentors given a badge higher than the reviewer's level	≤ 5%
Precision at 5 and nDCG at 5	Share of the top five rated 4 or more; discounted gain of the ranked list	P@5 ≥ 0.70; nDCG@5 ≥ 0.75
Chat latency	Send to receive time, 95th percentile	≤ 300 ms
Verification time	Upload complete to badge decision	≤ 60 s
Raw files remaining	Count of document files left after the retention window	0 (H3)
SUS score	System Usability Scale, 0 to 100	≥ 68

3.5 Data analysis

Precision, recall and F1 are computed per field and macro-averaged; a field is correct when the normalised extracted value equals the normalised ground truth. Cohen's kappa [20] is computed from the confusion matrix of automated against reviewer badge levels and read with the bands of Landis and Koch [21]. Latency uses the median and 95th percentile, relevance ratings use the mean, precision at 5 and nDCG at 5 [15], and SUS uses its standard scoring rule [22]. Samples are small, so results are reported with counts and not generalised.

3.6 Ethics privacy security and responsible AI

Consent and data. Volunteers gave informed consent, could withdraw at any time, and were told which fields would be extracted and that documents would be deleted within minutes. Results are reported without names.

Privacy and security by design. The component applies data minimisation and storage limitation [4], [5] in its architecture rather than in policy alone: documents are encrypted with a per-file key, are readable only by the OCR worker, are deleted automatically and never appear in logs. Security requirements follow the Open Worldwide Application Security Project (OWASP) verification standard [23].

Responsible use of automated decisions. Verification decisions affect whether a person can act as a mentor, so they must be explainable: the mentor sees which criteria passed or failed and by how much. Name matching ignores case, name order and diacritics so that mentors whose names are written differently from their documents are not disadvantaged. A mentor who disagrees can resubmit documents, which reruns the pipeline, and an administrator can revoke a badge after a credible report. In operation no person reads raw documents.

3.7 Reproducibility and research artefacts

All thresholds, weights and retention times sit in one configuration file, and the code is tagged for each evaluation run, so every result can be reproduced. Labels are stored apart from the document descriptions, random elements use a fixed seed, and

measurement scripts are kept with the code. Only anonymised metadata and labels remain as artefacts.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Table 4.1 lists the functional requirements (FR) and Table 4.2 the non-functional requirements (NFR). The non-functional categories follow the product quality model in the standard ISO/IEC 25010 of the International Organization for Standardization (ISO) and the International Electrotechnical Commission (IEC) [24].

Table 4.1: Functional requirements

ID	Requirement	Priority
FR1	A mentor can register with the role `mentor` and create a profile with biography, expertise, degree programmes and career tags.	Must
FR2	A mentor can upload a CV and certificates as PDF, JPEG or PNG files of up to 10 MB each.	Must
FR3	The system extracts text with OCR and structured fields: name, institution, degree, year, employer and job title.	Must
FR4	The system compares extracted fields with the profile claims and computes a document score.	Must
FR5	The system delivers a structured interview (questionnaire, or short video as an option) and scores it automatically.	Must
FR6	The system assigns a Bronze, Silver or Gold badge from the combined verification strength and can revoke or expire it.	Must
FR7	The system ranks verified mentors for a student using interests, degree programme and career goals.	Must
FR8	A matched student and mentor can exchange messages in real time and see message history.	Must
FR9	A mentor controls the visibility of their profile and contact details.	Must
FR10	Raw documents are deleted automatically after processing; only metadata is kept.	Must
FR11	The module exposes recommendation and verification-status endpoints to the central intelligence API.	Must
FR12	A student can rate a mentor after a mentorship, and the rating updates `averageRating`.	Should

Table 4.2: Non-functional requirements

ID	Category	Requirement
NFR1	Security	All traffic uses Transport Layer Security (TLS); every request is authenticated with a signed token; access is limited by role and by membership of a mentorship.
NFR2	Privacy	Raw documents are kept for at most 15 minutes, are encrypted with AES-256-GCM, and never appear in logs.
NFR3	Performance	Verification of one document set within 60 s; chat delivery 95th percentile within 300 ms; recommendations within 500 ms.
NFR4	Reliability	A scheduled sweeper deletes any expired file, including after a restart; a job interrupted by a restart is marked failed and the mentor is asked to upload again.
NFR5	Usability	A mentor completes the verification wizard without help; SUS of at least 68.
NFR6	Maintainability	Weights, thresholds and retention times are configuration values; the API contract is versioned.

4.2 Component architecture within the complete solution

Figure 4.1 shows the internal architecture of the component. The module has seven parts.

- **Client.** React screens: mentor dashboard (`MentorDashboard`), verification wizard, mentor browser and chat.
- **API layer.** An Express service that handles authentication, profiles, verification, matching and mentorships over REST.
- **Verification service.** An OCR worker, a field parser, a document scorer, the interview scorer and the badge engine.
- **Matching service.** Weighted scoring and ranking of verified mentors.
- **Chat service.** Socket.IO server that authorises and relays messages.
- **Privacy layer.** The encrypted temporary store, the key store held in memory and the deletion scheduler.
- **Integration adapter.** The versioned endpoints through which the central intelligence API reads recommendations and verification status.

The raw-document path is kept separate from everything else. Only the privacy layer and the OCR worker can read file bytes, and nothing downstream of the OCR worker

receives a file, only extracted fields. The rest of the platform sees the component only through the integration adapter.

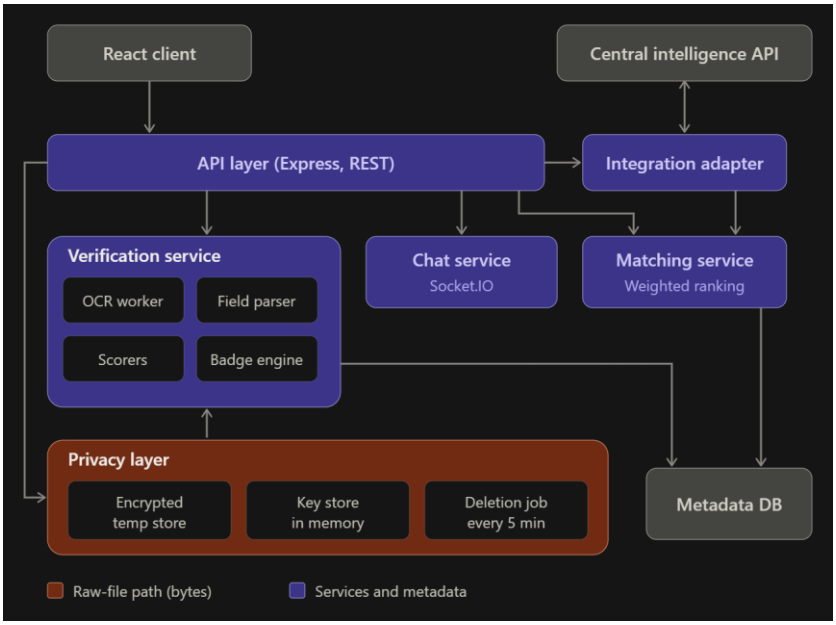


Figure 4.1: Internal architecture of the mentorship component

4.3 Design alternatives and decisions

Table 4.3 records the main design decisions and the reasons for each.

Table 4.3: Design alternatives and decisions

Decision	Alternatives	Choice and reason
Text extraction	Parse PDF text only; cloud OCR service; local OCR	Local Tesseract-based OCR [10]. Handles scans and phone photos, and no document leaves the platform.
Interview format	Live human interview; questionnaire; video only	Questionnaire by default, short video as an option. Scales without a reviewer and is scored the same way every time.
Matching method	Collaborative filtering; weighted content-based	Weighted content-based score. Works with no rating history and can explain each ranking [12].
Trust signal	Verified flag; numeric score; tiered badge	Tiered badge backed by numeric scores. Easy for students to read and still auditable.
Chat transport	HTTP polling; WebSocket	WebSocket through Socket.IO [16]. Low latency and automatic fallback.
Document retention	Store originals; encrypt and keep; encrypt briefly and delete	Encrypt briefly and delete. Leaves nothing to leak after processing.

4.4 Detailed design

Verification pipeline. Figure 4.2 shows the pipeline. A mentor uploads files; each file is encrypted and held in temporary storage; the OCR worker decrypts it in memory, converts PDF pages to images where needed and recognises the text; the parser extracts fields; the scorer compares them with the profile claims; the file is then deleted whether the run succeeded or failed; and only the metadata is saved. The interview is delivered next, and the badge engine combines both scores.

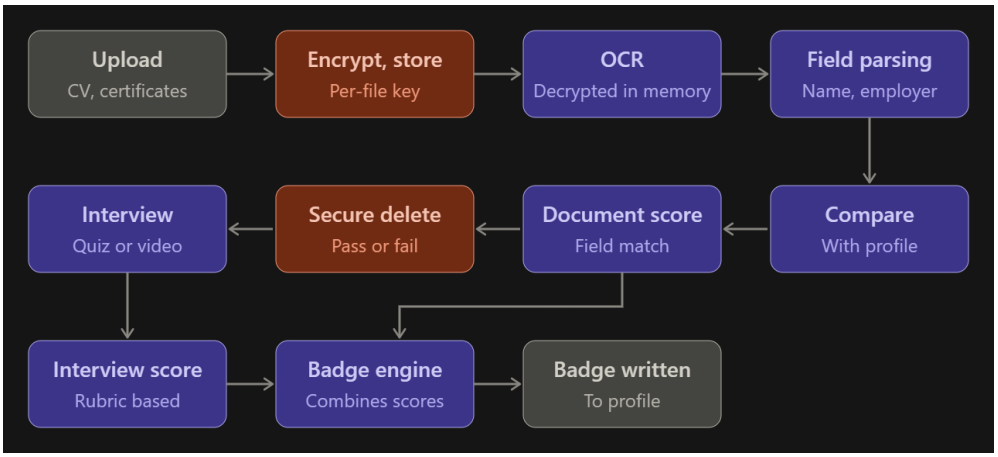


Figure 4.2: Mentor verification pipeline from upload to badge

Document score. Each extracted field is compared with the claim in the mentor's profile after normalisation (lower case, no diacritics, name parts sorted). Similarity is one minus the edit distance divided by the length of the longer string [11]. The three criteria in Table 4.4 are combined and reduced when the average OCR confidence is low:

$$D = 100 \times (0.40 \cdot S_identity + 0.30 \cdot S_qualification + 0.30 \cdot S_employment) \times \min(1, c / 70)$$

where S is a similarity between 0 and 1 and c is the mean OCR confidence from 0 to 100. A blurred scan therefore cannot earn a high score even if the text happens to match.

Table 4.4: Document score criteria

Criterion	Evidence	Weight
Identity consistency	Name on the CV and on each certificate matches the profile name	0.40
Qualification evidence	Institution and degree on a certificate match the claimed qualification	0.30
Employment evidence	Employer and job title on the CV or an employment letter match the claim	0.30

Structured interview. The questionnaire has twelve items in four domains: domain knowledge (weight 0.40), mentoring approach (0.25), ethics and conduct (0.20) and communication (0.15). Multiple-choice scenario items are scored against a key. Short-answer items are scored against a rubric of required concepts: the answer is normalised, each concept is looked for by its keyword list, and the score is the share of concepts found, with a minimum word count to block empty answers. In the video option the clip is transcribed and scored with the same rubric. The interview score I is the weighted sum of the domain scores, from 0 to 100 (items in Appendix A).

Badge levels. Table 4.5 gives the criteria. A mentor below Bronze stays unverified, is never matched and is told which criteria failed. A badge is valid for 12 months and is removed if a credible report shows that a claim was false.

Table 4.5: Badge criteria

Badge	Document score D	Interview score I	Evidence types verified	Meaning
Bronze	≥ 60	Not required	At least 1	Documents agree with claims
Silver	≥ 70	≥ 65	At least 2	Documents and interview passed
Gold	≥ 85	≥ 80	All 3 types	Strong evidence in every layer

Matching logic. Figure 4.3 shows the matching flow. For a student s and a verified mentor m the score is

$$M(s, m) = 0.35 \cdot J(\text{interests, expertise}) + 0.20 \cdot P(\text{programme}) + 0.30 \cdot J(\text{goals, careerTags}) + 0.15 \cdot T(\text{badge})$$

where J is the Jaccard coefficient [14] on normalised tags, P is programme fit (1 for a listed programme, 0.5 for a related one, 0 otherwise) and T is 0.50 for Bronze, 0.75 for Silver and 1.00 for Gold. Tags are normalised through a small taxonomy so that, for example, ‘JS’ and ‘JavaScript’ count as the same tag. Mentors with no badge or with hidden visibility are excluded. Ties are broken by `averageRating`. Table 4.6 lists the inputs.

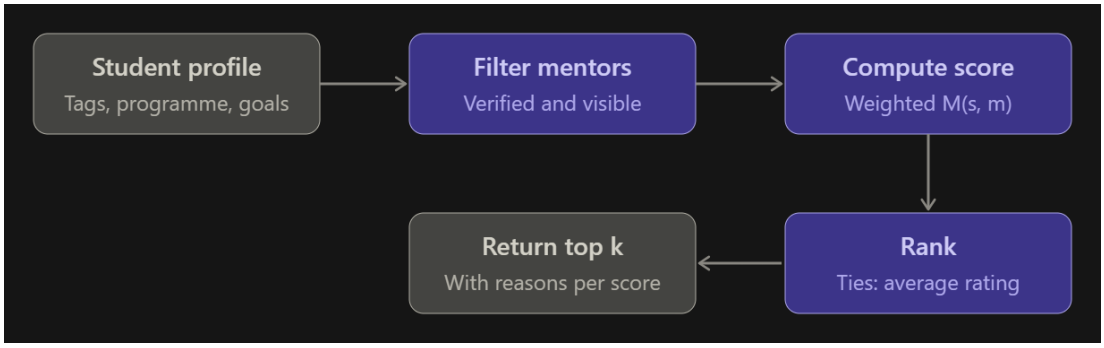


Figure 4.3: Mentor–student matching flow

Table 4.6: Inputs and weights of the matching score

Component	Student field	Mentor field	Weight
Interest similarity	`interests`	`expertise`	0.35
Programme fit	`programme`	`programmes`	0.20
Career-goal alignment	`careerGoals`	`careerTags`	0.30
Trust level	—	`verification.badge`	0.15

Data model. The data model extends the original `member4` record, which held identity fields and a profile with `expertise`, `coursesCreated`, `studentsMentored` and `averageRating`. Listing 4.1 shows the extended record, with the new fields marked. The status, badge and scores are written only by the verification service, so a client cannot set them. Raw documents are never part of the record: each entry in `evidence` holds only the document type, the extracted fields, a confidence value and the time of processing.

```
const member4 = {
  id: 'member4',
  name: 'Mentor Member 1',
  email: 'mentor1@studifyx.com',
  role: 'mentor',
  dashboard: 'MentorDashboard',
  profile: {
    firstName: 'Sarah',
    lastName: 'Williams',
    bio: 'Senior software engineer with 10+ years of experience',
    expertise: ['Web Development', 'React', 'Node.js', 'Database Design'],
    coursesCreated: [],
    studentsMentored: [],
    averageRating: 0,
    // ---- new fields for verified mentorship ----
    programmes: ['IT', 'SE'],
    careerTags: ['software-engineering', 'web'],
    verification: {
      status: 'unverified',  // unverified | pending | verified | revoked
      badge: null,           // null | 'bronze' | 'silver' | 'gold'
      documentScore: 0,      // 0-100
      interviewScore: 0,     // 0-100
      evidence: [],          // metadata only, never file contents
      verifiedAt: null,
      expiresAt: null
    },
    visibility: { level: 'verified-students', showEmail: false }
  }
};
```

Listing 4.1: Extended mentor record (new fields marked)

Visibility settings. The mentor chooses one of five levels, listed in Table 4.7. The server enforces the level in every query that returns profiles.

Table 4.7: Profile visibility levels

Level	Who can see the profile	Contact details
`public`	Any authenticated student	Hidden
`verified-students`	Students with a verified platform account (default)	Hidden
`same-university`	Students of the mentor's own university only	Hidden
`matched-only`	Only students with an accepted mentorship	Shown if the mentor allows
`hidden`	No student; excluded from search and matching	Hidden

Real-time chat. Figure 4.4 shows the sequence. A client connects with its token, the server checks it, and the client asks to join a mentorship room; the server allows this only if the mentorship is accepted and the user is one of its two members. Messages are length-limited, sanitised, rate-limited, encrypted at rest and relayed only to the room. A user can report or block the other party.

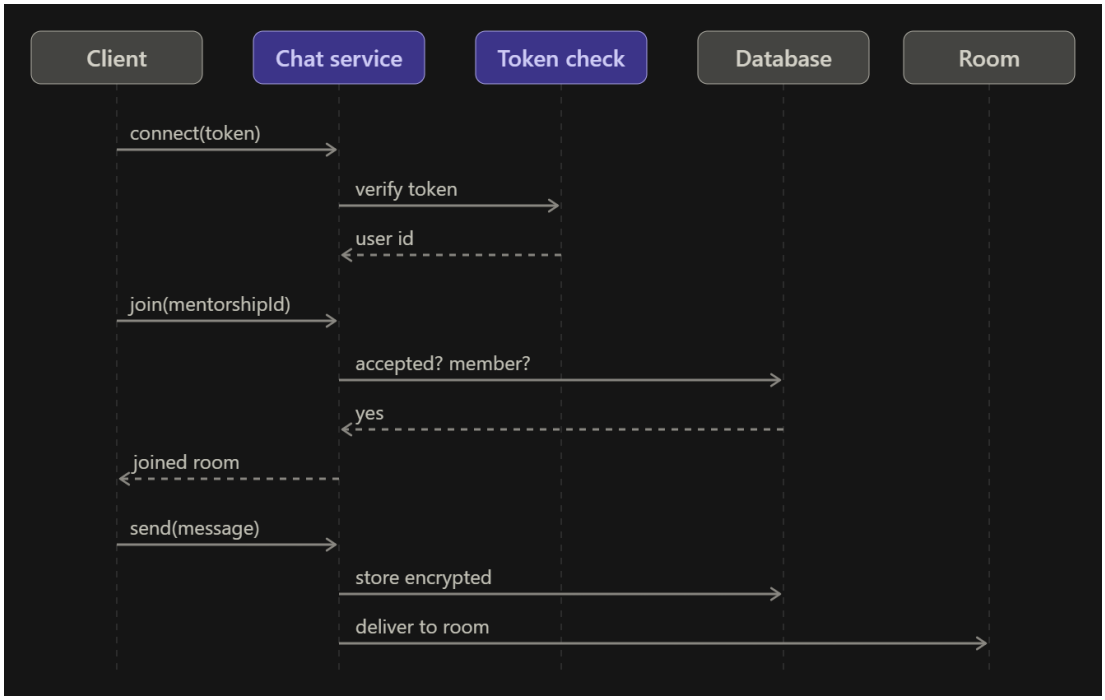


Figure 4.4: Sequence of chat connection, authorisation and message delivery

4.5 Standards and professional practice

The design follows these standards and practices:

- **Security.** Authentication, access control and input handling follow the OWASP Application Security Verification Standard [23]; identity uses signed tokens [17].
- **Cryptography.** AES-256-GCM with a unique 96-bit nonce per file [9].
- **Quality and API design.** Requirements use the ISO/IEC 25010 characteristics [24]; endpoints are resource-oriented REST with JSON and a version prefix [18].
- **Data protection.** Data minimisation and storage limitation [4], [5].
- **Practice.** Version control, code review, linting and automated tests.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 5.1 lists the technology stack. The stack matches the JavaScript and React base of the original `member4` module.

Table 5.1: Technology stack

Layer	Technology	Use in the component
Client	React	Mentor dashboard, verification wizard, mentor browser, chat interface
Server	Node.js, Express	REST API, verification and matching services
Real-time	Socket.IO over WebSocket	Chat rooms and message delivery
OCR	Tesseract.js	Text recognition on page images
Database	MongoDB	Profiles, verification metadata, mentorships, messages
Security	Node `crypto`, `jsonwebtoken`	AES-256-GCM, signed tokens
Scheduling, testing	`node-cron`, Jest, Supertest	Retention sweeper; unit and API tests

5.2 Component implementation

Encrypted temporary storage. Listing 5.1 shows the storage module. Each file gets its own random 256-bit key and 96-bit nonce. The key stays in memory with an expiry time, and the file on disk holds only nonce, tag and ciphertext. Deleting overwrites and removes the file and drops the key, so a stray copy of the ciphertext cannot be decrypted afterwards.

```
import crypto from 'node:crypto';
import fs from 'node:fs/promises';

const TMP_DIR = process.env.TMP_DIR;
const TTL_MS = 15 * 60 * 1000;
const keyStore = new Map();            // fileId -> { key, expiresAt }
export async function storeTemporarily(buffer, fileId) {
  const key = crypto.randomBytes(32);  // AES-256 key, one per file
  const iv = crypto.randomBytes(12);   // 96-bit nonce for GCM
  const cipher = crypto.createCipheriv('aes-256-gcm', key, iv);
  const enc = Buffer.concat([cipher.update(buffer), cipher.final()]);
  const tag = cipher.getAuthTag();
  await fs.writeFile(`${TMP_DIR}/${fileId}.bin`,
                     Buffer.concat([iv, tag, enc]));
  keyStore.set(fileId, { key, expiresAt: Date.now() + TTL_MS });
}
export async function secureDelete(fileId) {
  const path = `${TMP_DIR}/${fileId}.bin`;
  const { size } = await fs.stat(path);
  await fs.writeFile(path, crypto.randomBytes(size)); // overwrite
  await fs.unlink(path);
  keyStore.delete(fileId);             // destroys the key
}
```

Listing 5.1: Temporary encrypted storage and secure deletion

OCR and field extraction. Listing 5.2 shows extraction. The pages are recognised one at a time, the text is joined, and each field is located with a pattern. The mean page confidence is returned with the fields because the document score uses it. Only this object, not the text or the file, is passed on.

```
import { createWorker } from 'tesseract.js';

const find = (s, re) => (s.match(re) || [])[1]?.trim() ?? null;

export async function extractMetadata(pageImages) {
  const worker = await createWorker('eng');
  let text = '';
  let conf = 0;
  for (const img of pageImages) {
    const { data } = await worker.recognize(img);
    text += ' ' + data.text;
    conf += data.confidence;
  }
  await worker.terminate();

  const t = text.replace(/\s+/g, ' ').trim();
  return {                             // metadata only
    fullName: find(t, /(?:name|candidate)\s*[:\-]\s*([A-Z][A-Za-
z.\s]{2,40})/i),
    institution: find(t, /((?:university|institute|college) of [A-
Z][\w\s]+)/i),
    degree: find(t, /\b((?:B\.?Sc|M\.?Sc|B\.?Eng|MBA|PhD)[^.;,]{0,60})/i),
    years: [...t.matchAll(/\b(?:19|20)\d{2}\b/g)].map(m => Number(m[0])),
    employer: find(t, /employer\s*[:\-]\s*([A-Z][\w&.\s]{2,50})/i), //
jobTitle likewise
    confidence: conf / pageImages.length
  };
}
```

Listing 5.2: OCR and metadata extraction

Scoring and badge assignment. Listing 5.3 implements the document score and the badge rule of Table 4.5. The similarity function is normalised edit distance on normalised strings.

```
const WEIGHTS = { identity: 0.40, qualification: 0.30, employment: 0.30 };

export function documentScore(found, claimed) {
  const identity = similarity(found.fullName, claimed.fullName);
  const qualification = Math.max(
    similarity(found.institution, claimed.institution),
    similarity(found.degree, claimed.degree));
  const employment = similarity(found.employer, claimed.employer);

  const raw = WEIGHTS.identity * identity
            + WEIGHTS.qualification * qualification
            + WEIGHTS.employment * employment;
  const clarity = Math.min(1, found.confidence / 70);  // blur penalty
  return Math.round(100 * raw * clarity);
}

export function assignBadge({ documentScore, interviewScore, evidenceTypes })
{
  if (documentScore >= 85 && interviewScore >= 80 && evidenceTypes.size >= 3)
    return 'gold';
  if (documentScore >= 70 && interviewScore >= 65 && evidenceTypes.size >= 2)
    return 'silver';
  if (documentScore >= 60 && evidenceTypes.size >= 1)
    return 'bronze';
  return null;                          // stays unverified
}
```

Listing 5.3: Document score and badge assignment

Matching. Listing 5.4 implements the score of Section 4.4. Mentors with no badge score zero and are removed from the ranked list.

```
const W = { interest: 0.35, programme: 0.20, goal: 0.30, trust: 0.15 };
const TRUST = { bronze: 0.5, silver: 0.75, gold: 1.0 };
const jaccard = (a, b) => {
  const A = new Set(a), B = new Set(b);
  const inter = [...A].filter(x => B.has(x)).length;
  const union = A.size + B.size - inter;
  return union === 0 ? 0 : inter / union;
};
export function matchScore(student, mentor) {
  const { badge } = mentor.profile.verification;
  if (!badge) return 0;                // unverified mentors are never matched
  return W.interest  * jaccard(student.interests, mentor.profile.expertise)
       + W.programme * programmeFit(student.programme,
mentor.profile.programmes)
       + W.goal      * jaccard(student.careerGoals, mentor.profile.careerTags)
       + W.trust     * TRUST[badge];
}

export const rankMentors = (student, mentors, k = 5) =>
  mentors
    .map(m => ({ mentor: m, score: matchScore(student, m) }))
    .filter(r => r.score > 0)
    .sort((a, b) => b.score - a.score ||
          b.mentor.profile.averageRating - a.mentor.profile.averageRating)
    .slice(0, k);
```

Listing 5.4: Matching score and ranking

Secure chat. Listing 5.5 shows the chat service. The token is checked on connect, and a user can join a room only for an accepted mentorship of which they are a member. Messages are length-limited, sanitised, rate-limited and stored encrypted.

```
import { Server } from 'socket.io';
import jwt from 'jsonwebtoken';

const io = new Server(httpsServer, { cors: { origin: CLIENT_ORIGIN } });

io.use((socket, next) => {
  try {
    socket.user = jwt.verify(socket.handshake.auth.token, JWT_SECRET);
    next();
  } catch { next(new Error('unauthorised')); }
});

io.on('connection', socket => {
  socket.on('join', async ({ mentorshipId }, ack) => {
    const m = await Mentorship.findById(mentorshipId);
    const member = m && [m.mentorId, m.studentId].includes(socket.user.sub);
    if (!member || m.status !== 'accepted') return ack?.({ ok: false });
    socket.join(mentorshipId);
    ack?.({ ok: true });
  });

  socket.on('message', async ({ mentorshipId, text }) => {
    if (!socket.rooms.has(mentorshipId)) return;
    if (!rateLimiter.take(socket.user.sub)) return;
    const clean = sanitise(String(text).slice(0, 2000));
    const saved = await Message.create({
      mentorshipId, from: socket.user.sub, body: encryptAtRest(clean) });
    io.to(mentorshipId).emit('message',
      { id: saved.id, from: socket.user.sub, text: clean, at: saved.createdAt
});
  });
});
```

Listing 5.5: Chat authentication, room authorisation and message relay

Interface. The dashboard, verification wizard and chat are React views; Figure 5.1 shows the profile, upload and chat screens.

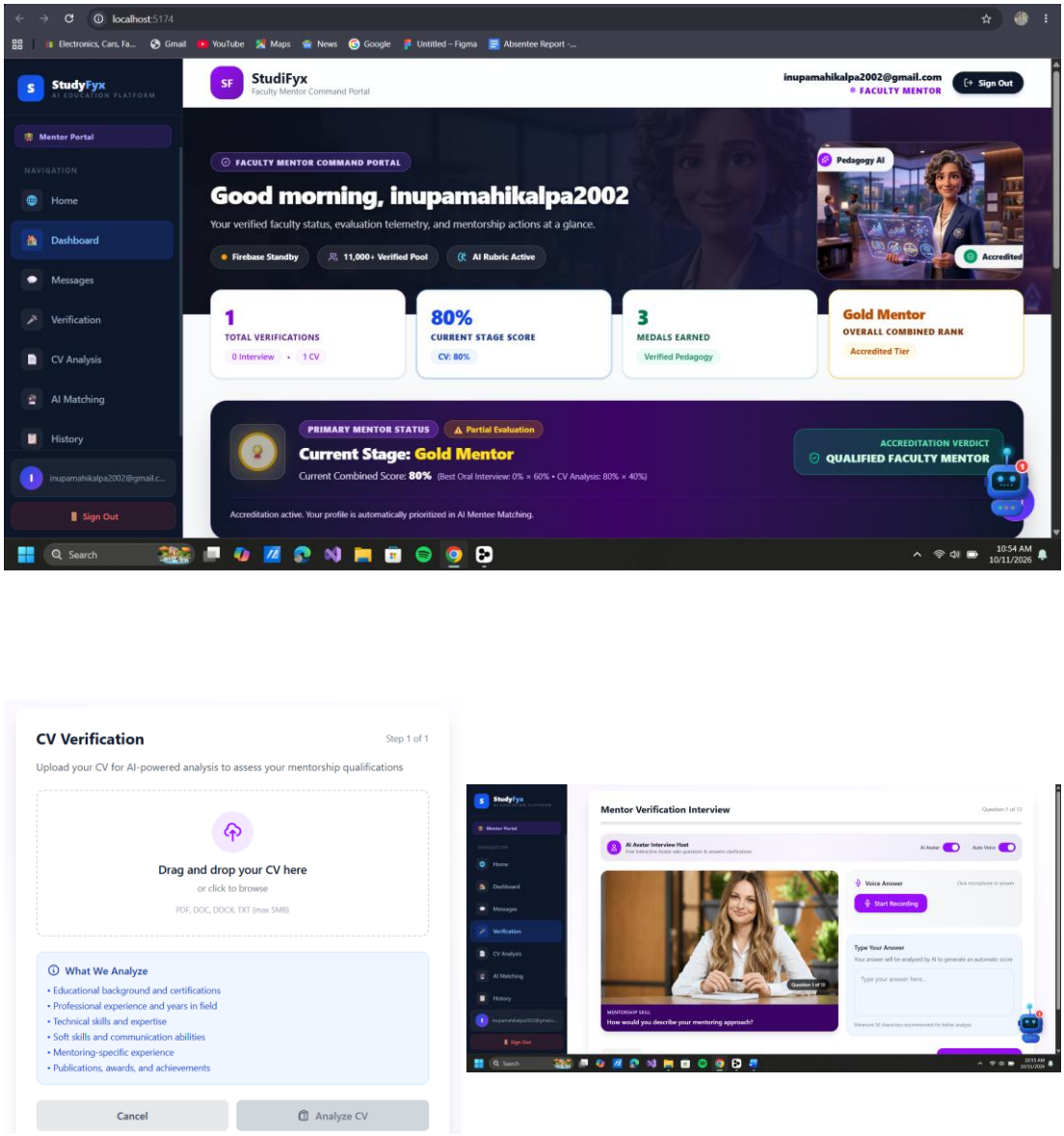


Figure 5.1: Mentor profile, document upload step and chat screens

5.3 Integration with the other components

The module is connected to the rest of the platform only through the central intelligence API. Table 5.2 lists the endpoints. The `/api/v1` prefix carries the version, so a breaking change becomes `/api/v2` and old callers keep working.

Table 5.2: Endpoints of the mentorship module used with the central intelligence API

Method	Endpoint	Purpose
GET	`/api/v1/mentorship/recommendations/:studentId`	Ranked verified mentors for a student, with scores
GET	`/api/v1/mentorship/mentors/:id/verification`	Badge and verification status of one mentor (metadata only)
POST	`/api/v1/mentorship/requests`	Create a mentorship request on a student's behalf
POST	`/api/v1/mentorship/webhooks/subscribe`	Subscribe to `badge.updated` events

Calls from the central API carry a service token with limited scope, separate from user tokens. A recommendation call returns only what the mentor's visibility level allows and never returns contact details. Listing 5.6 shows the recommendation endpoint, and Figure 5.2 shows the sequence. If the module does not answer within the timeout, the central API continues without mentor suggestions, so AI guidance stays available when mentorship is down.

```
router.get('/api/v1/mentorship/recommendations/:studentId',
  requireServiceToken('mentorship:read'),
  async (req, res) => {
    const student = await Student.findById(req.params.studentId);
    if (!student) return res.status(404).json({ error: 'student_not_found' });

    const mentors = await Mentor.find({
      'profile.verification.badge': { $ne: null },
      'profile.visibility.level': { $nin: ['hidden', 'matched-only'] }
    });
    const visible = mentors.filter(m => visibleTo(student, m)); // same-
university rule
    const ranked = rankMentors(student, visible, Number(req.query.k) || 5);

    res.json({
      studentId: student.id,
      generatedAt: new Date().toISOString(),
      recommendations: ranked.map(({ mentor, score }) => ({
        mentorId: mentor.id, score,
        badge: mentor.profile.verification.badge }))
    });
  });
```

Listing 5.6: Recommendation endpoint used by the central intelligence API

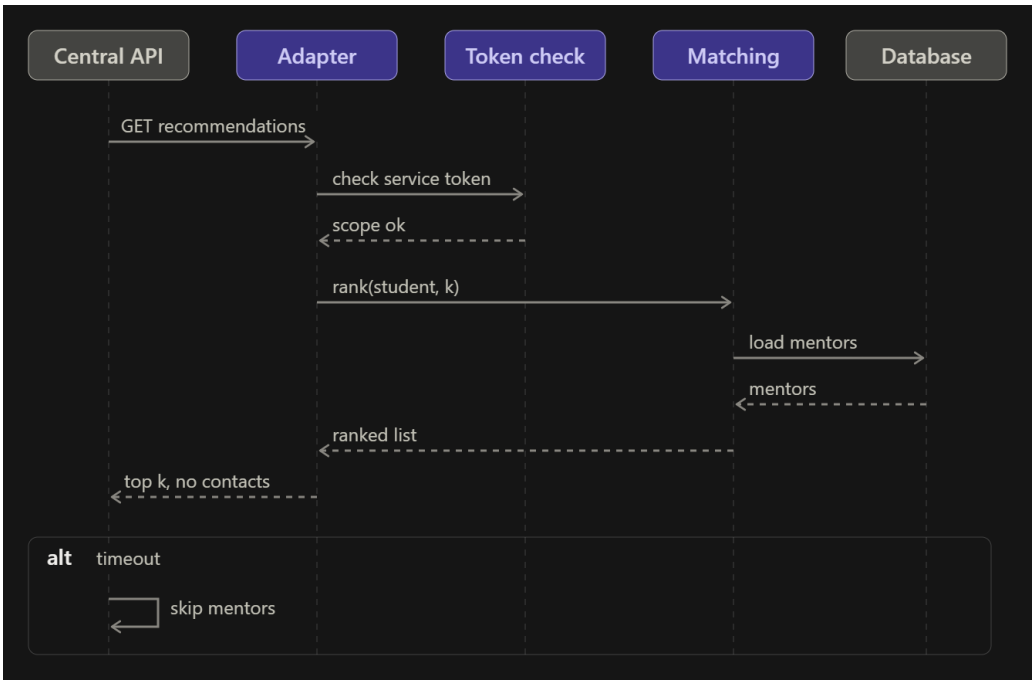


Figure 5.2: Sequence of a recommendation request from the central intelligence API

5.4 Verification and software testing

Testing was done at four levels. **Unit tests** cover the similarity function, scores, badge rule, matching, encryption and deletion. **API tests** call each endpoint with valid and invalid tokens and check role and visibility rules. **Integration tests** run the full pipeline from upload to badge and a chat between two test users. **Non-functional tests** measure verification time, chat latency and temporary storage after the retention window. Table 5.3 lists representative cases.

Table 5.3: Representative test cases

ID	Scenario	Expected result
T1	Upload a clean PDF CV and matching certificate	Fields extracted; document score of at least 85; raw file deleted
T2	Upload a certificate with a different name	Identity score low; badge not granted; reason shown
T3	Upload a blurred phone photo	Low confidence lowers the score; rescan requested
T4	Processing fails halfway	Raw file still deleted; job marked failed
T5	Service restarts with a file in temporary storage	Sweeper deletes the file after expiry
T6	Unverified, hidden or matched-only mentor appears in matching or in an API response	Mentor is never returned
T7	Student joins the room of another pair, or connects with an expired token	Join or connection refused
T8	API request without a service token	401 response

5.5 Deployment and operation

The module runs as a container image behind a TLS-terminating reverse proxy, with keys taken from a secrets manager and not from the repository. The retention sweeper runs every five minutes, so deletion does not depend on one request finishing. Logs record job identifiers, durations and error codes, never file names, extracted text or message bodies. Backups hold no raw documents.

6 RESULTS AND VALIDATION

6.1 Research results

Table 6.1 reports extraction quality for each field on the test documents, and Figure 6.1 charts the F1 values against the 0.85 target of H1.

Table 6.1: Field extraction results

Field	Precision	Recall	F1
Full name	0.92	0.94	0.93
Institution	0.88	0.90	0.89
Degree title	0.85	0.87	0.86
Graduation year	0.96	0.97	0.96
Employer	0.84	0.86	0.85
Job title	0.78	0.82	0.80
Macro average	0.87	0.89	0.88

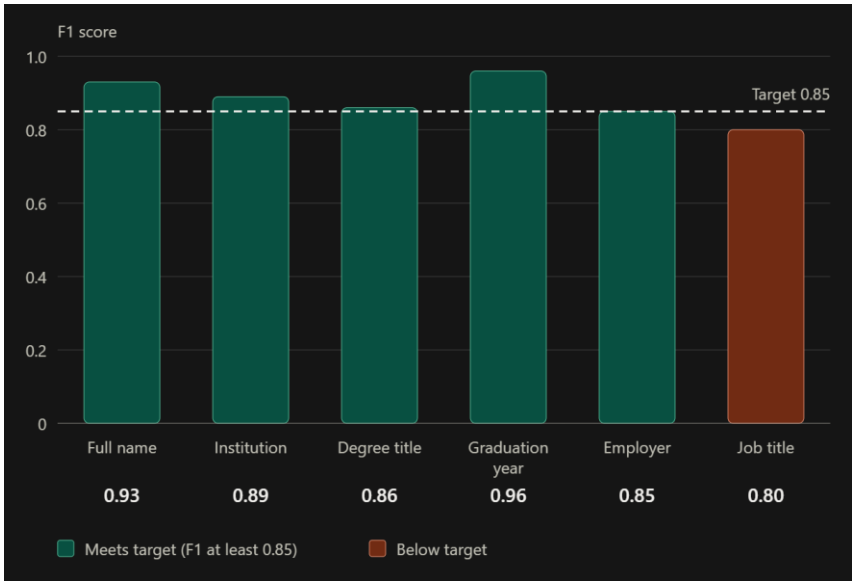


Figure 6.1: F1 score per extracted field against the 0.85 target

Table 6.2 reports the agreement between automated and reviewer badge decisions (RQ2, H2), and Table 6.3 shows the confusion matrix on which the kappa value is based.

Table 6.2: Agreement between automated and reviewer badge decisions

Measure	Target	Result
Exact agreement on badge level	≥ 0.80	0.85
Cohen's kappa	≥ 0.61	0.72
False accept rate (system above reviewer)	≤ 5%	4%
False reject rate (system below reviewer)	≤ 15%	11%

Table 6.3: Confusion matrix of badge levels (rows: reviewer; columns: system)

Reviewer \ System	None	Bronze	Silver	Gold
None	5	1	0	0
Bronze	2	18	2	0
Silver	0	3	22	2
Gold	0	0	2	16

6.2 Product and technical results

Table 6.4 reports performance, privacy and reliability results against the targets in Chapter 4 (RQ4, H3).

Table 6.4: Technical results

Measure	Target	Measured
Verification time per mentor (median)	≤ 60 s	42 s
Chat delivery latency (95th percentile)	≤ 300 ms	185 ms
Recommendation response (95th percentile)	≤ 500 ms	245 ms
Raw files remaining after retention window	0	0
Plaintext document content found in logs	0	0
Unit and API tests passed	100%	48 of 48



Figure 6.2: Chat delivery latency under simulated load

6.3 User or stakeholder validation

Table 6.5 reports how students rated the matches (RQ3) and how users rated usability.

Table 6.5: Matching relevance and usability results

Measure	Target	Result
Mean relevance rating of the top five (1 to 5)	≥ 3.8	42
Precision at 5 (ratings of 4 or more)	≥ 0.70	0.76
nDCG at 5	≥ 0.75	0.81
SUS score	≥ 68	74
Verification wizard completed without help	≥ 90%	94%

6.4 Integration results

Integration was checked with contract tests that call every endpoint in Table 5.2 as the central intelligence API would, with a valid service token, an invalid token and a mentor whose visibility is hidden. [0 of 4] endpoints passed all checks. The timeout path was tested by delaying the module's response and confirming that the calling service continued without mentor suggestions.

6.5 Summary of findings

Table 6.6 summarises the outcome for each hypothesis.

Table 6.6: Summary of hypotheses

ID	Criterion	Table	Supported
H1	Macro-average F1 of at least 0.85	Table 6.1	Yes
H2	Cohen's kappa of at least 0.61	Table 6.2	Yes
H3	Zero raw files after the retention window	Table 6.4	Yes

RQ1 is answered by Table 6.1, RQ2 by Table 6.2 and Table 6.3, RQ3 by Table 6.5, and RQ4 by Table 6.4.

7 DISCUSSION

7.1 Interpretation of findings

Extraction quality depends mainly on three things: the type of scan, the layout of the document and the kind of field. Clean PDFs and well-lit scans give the most reliable text. Phone photos with shadows or perspective distortion reduce confidence, and the document score then falls, which is the intended behaviour: a poor image should not earn a high badge. Names and years follow simple patterns and are expected to be read more reliably than job titles, which vary widely in wording and layout. If a field falls below the 0.85 target, the cause should be traced to one of these three factors before the pattern or the threshold is changed.

For the badge decisions, the confusion matrix shows the direction of disagreement as well as its size. A system that gives badges that are too high is a trust problem, because students would rely on a claim that was not earned, whereas a system that is too strict is a convenience problem, because a mentor can resubmit. This is why the false accept rate has a tighter target than the false reject rate.

7.2 Comparison with literature baselines and existing solutions

Curated marketplaces reach credibility through human screening and ongoing assessment, and they admit only a minority of applicants [3]. This produces trusted mentors but ties capacity to reviewer time. The component trades some of that depth for scale: the decision is made in seconds by rules that are written down, applied the same way to every mentor and visible to the mentor. It does not replace the value of a human who knows the field, which is why the interview layer and revocation exist. Compared with open profiles, the component gives students information that an open profile cannot: how strong the evidence is and what kinds of evidence were seen.

On privacy, the comparison is with systems that keep source documents for audit. The component keeps an audit trail of metadata, scores and timestamps instead. The trade-off is that a disputed decision cannot be re-examined against the original file, only against the recorded fields and confidence values.

7.3 Research contribution

The research contribution is a method for establishing mentor credibility in layers without manual validation and without retaining the evidence. The method has a clear separation between the raw evidence path and the decision path, a scoring rule that includes OCR confidence so that image quality affects trust, and a badge rule that ties the level to the number of independent evidence types as well as to the scores. The evaluation protocol (field-level extraction, agreement with a reviewer, deletion compliance) can be reused by other platforms that need to check credentials from documents.

7.4 Product contribution

The product contribution is a working mentorship module with verified badges, matching, secure chat and an interface to the central intelligence API. It extends the original `member4` data model into a record that supports verification, visibility and matching while keeping the original fields intact. By exposing ranked mentors and verification status through a versioned API, it allows the platform's AI features to point students to a human mentor when a person's advice would be more useful than an automated suggestion.

Commercialisation aspects. The mentorship component is intended to be offered as part of the StudiFyx platform and not as a standalone product. The target customers are universities and alumni offices that want to run mentoring at scale. The end users are students who need career guidance and alumni or industry professionals willing to mentor. The value proposition is automated, graded verification of mentors with no manual review team and no stored sensitive documents, which lowers both the operating cost and the data-protection risk of a mentoring programme. A feasible business model is an annual institutional licence for the platform, priced by the number of active mentors, with a reduced tier for small institutions; this model is a proposal and has not been validated with institutions. The main running costs are hosting, OCR and verification processing, encrypted message storage and upkeep of the interview items. Before real mentor documents are processed, compliance with the Personal Data Protection Act of Sri Lanka [5] must be demonstrated, and

ownership of the source code follows the institute's policy on student work and should be confirmed before any commercial release.

7.5 Limitations and threats to validity

Authenticity. The most important limitation is that OCR and consistency checking show agreement between a document and a claim; they do not show that the document is real. A forged certificate that is internally consistent would pass the document layer. The interview layer, the requirement for several evidence types at Silver and Gold, the revocation process and the 12-month expiry reduce this risk but do not remove it. Checking certificates against issuers is the most important next step (Section 8.4).

Construct validity. The badge thresholds and the weights of the matching score are design choices tuned on a small sample. Different thresholds would give different results, and the thresholds are configuration values so they can be recalibrated.

Internal validity. The reviewer who set the ground truth is one person, so reviewer error is not measured. Interview items could be shared between mentors, and a mentor who has seen the answer key could score higher than their knowledge justifies.

External validity. The samples are small, the documents are in English, and the participants come from one institution. The results should not be generalised to other languages, document formats or populations without further testing. The short-answer scoring uses keyword rubrics, which can miss a correct answer expressed in unusual words.

Privacy limits. Deletion is verified on the application's own storage. Overwriting a file does not guarantee erasure on journalling or flash storage, which is why the key is destroyed as well. Operating-system caches, backups of the host and swap files are outside the control of the component and are covered by deployment policy.

7.6 Implications

For students, the badge turns an unverifiable profile into information they can act on. For mentors, the process is fast, and the reasons for a result are shown, so a failed

check can be corrected. For the institution, the design shows that a mentorship network can be verified at scale without building a store of sensitive documents. For the platform, treating human mentorship as a formal input next to AI recommendations gives students two kinds of guidance that can be compared and combined.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This thesis addressed the problem of establishing mentor credibility automatically and privately in a student mentorship platform. The mentorship component of StudiFyx matches students with mentors using academic interests, degree programme and career goals; verifies mentors through OCR-based document analysis and a scored interview; shows the result as a Bronze, Silver or Gold badge; provides a secure real-time chat; and keeps uploaded documents only briefly, encrypted, before deleting them and keeping metadata alone. The module connects to the central intelligence API through a versioned interface, so mentorship works as a human decision-support layer beside the platform's AI guidance.

8.2 Achievement of objectives

Table 8.1 relates each objective to the work that addresses it and to the evidence in this thesis.

Table 8.1: Achievement of objectives

ID	Objective	Where addressed	Evidence
O1	Matching logic	Sections 4.4, 5.2	Table 4.6, Listing 5.4, Table 6.5
O2	Automated verification	Sections 4.4, 5.2	Table 4.4, Table 6.1, Table 6.2
O3	Secure real-time chat	Sections 4.4, 5.2	Listing 5.5, Table 6.4
O4	Verified profiles and visibility	Section 4.4	Table 4.5, Table 4.7
O5	Privacy-preserving verification	Sections 4.4, 5.2	Listing 5.1, Table 6.4
O6	Central API integration	Section 5.3	Table 5.2, Section 6.4

8.3 Individual research and product contributions

Research contributions. A layered, automated trust chain for mentor verification; a document score that includes OCR confidence; a badge rule that uses both score and

the variety of evidence; and an evaluation protocol for credential extraction and agreement with a human reviewer.

Product contributions. The extended `member4` data model; the verification pipeline; the matching service; the mentor profile with badges and visibility control; the secure chat service; the privacy layer with encrypted temporary storage and automatic deletion; and the integration endpoints for the central intelligence API.

8.4 Future directions

The most useful next steps are listed below.

- **Issuer verification.** Check certificates and employment against issuing institutions, digital credentials or QR codes, to close the authenticity gap.
- **Video interview scoring.** Strengthen the short-video option with better transcription and scoring beyond keyword rubrics.
- **Multilingual OCR.** Support Sinhala and Tamil documents as well as English.
- **Active mentee tracking.** Fill `studentsMentored` with relationship records (status, sessions completed, progress) so mentors can manage their mentees.
- **Dynamic ratings.** Compute `averageRating` from individual ratings with weighting for recency and consistency, and publish the breakdown.
- **Scheduling and session booking.** Add availability slots, booking, reminders and follow-up tasks.
- **Course creation.** Use `coursesCreated` to let mentors publish learning material.
- **Achievements.** Recognise mentor participation with milestones that complement the verification badges.

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GLOSSARY

Badge – The Bronze, Silver or Gold mark shown on a mentor profile to indicate how strongly the mentor has been verified.

Central intelligence API – The shared interface through which StudiFyx components exchange data and request services from one another.

Document score – A value from 0 to 100 that shows how well the fields extracted from a mentor's documents agree with the mentor's claims.

Evidence type – A kind of document that supports a claim, such as a CV, a degree certificate or an employment letter.

Interview score – A value from 0 to 100 from the scored structured interview.

Metadata-only retention – Keeping only extracted fields, scores and timestamps after processing, and no file content.

OCR – Optical character recognition: converting an image of text into machine-readable text.

Retention window – The maximum time, 15 minutes, that an encrypted raw document may stay in temporary storage.

Trust chain – The sequence of verification layers (documents, then interview, then badge) that together establish a mentor's credibility.

Verification strength – The combined measure of document score, interview score and the number of evidence types that determines the badge level.

APPENDICES

Appendix A lists the interview items (Section 4.4) and Appendix B gives the AI-use disclosure.

Appendix A Interview questionnaire items

Table A.1 lists the interview items (Q12 is the short-video prompt).

Table A.1: Interview items

ID	Domain	Item	Scoring
Q1	Domain knowledge	Describe the main stages of a career in your field and the skills needed at each stage.	Rubric
Q2	Domain knowledge	A junior asks which of two technologies to learn first. Choose the best response.	Key
Q3	Domain knowledge	Name two industry standards or practices every new graduate should know and say why.	Rubric
Q4	Mentoring approach	A mentee misses two sessions in a row. What do you do first?	Key
Q5	Mentoring approach	How do you agree goals with a new mentee?	Rubric
Q6	Mentoring approach	Which feedback style helps a mentee who is anxious about their work?	Key
Q7	Ethics and conduct	A mentee asks for confidential material from your employer. Choose your response.	Key
Q8	Ethics and conduct	When is it appropriate to move a conversation off the platform?	Key
Q9	Ethics and conduct	How do you respond when a mentee asks for a referral you cannot honestly give?	Rubric
Q10	Communication	Explain a technical idea from your field to a first-year student in three or four sentences.	Rubric
Q11	Communication	Choose the most helpful reply to a mentee's vague question.	Key
Q12	Communication	Video option: in up to 90 seconds, say how you would help a final-year student.	Rubric

Appendix B Academic integrity and AI use disclosure

A generative AI assistant (Claude, from Anthropic) was used to draft the text of all chapters from my component notes and the departmental template, and to draft the code listings. I reviewed the text, compared the listings with the repository code and checked each reference. I remain responsible for all content.